

ONCRAG

A Vectorless Retrieval-Augmented Generation System for Oncology Question Answering

FINAL EVALUATION REPORT

Three-Way Ablation Study — Full LAQA System · No-LAQA RAG · No-RAG (LLM-only) Baseline

200 Oncology Questions | S.C.O.P.E.E Framework | Paired t-Tests | 95% Confidence Intervals

Prepared by **Roopa** · B.Tech CS & Data Science, Sai University · AI/ML Research Internship, NITPY · July 2026

All generation configurations use Mistral-7B-Instruct-v0.2 (4-bit NF4 quantized). Faithfulness and S.C.O.P.E.E scores are judged by an independent evaluator model, Llama3-Med42 (HuggingFace), not the generation model itself.

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Abstract

Large language models deployed for clinical question answering are prone to hallucination — the generation of fluent statements unsupported by any verifiable source — a failure mode with particular consequence in oncology. This report presents OncRAG, a vectorless-first, agentic retrieval-augmented generation (RAG) system for oncology question answering, and evaluates it through a controlled three-way ablation against two reduced configurations: a No-RAG baseline that generates from parametric knowledge alone, and a No-LAQA variant that performs hybrid retrieval without query-aware preprocessing. OncRAG combines rule-based agentic chunking, BM25 and FAISS hybrid retrieval fused via Reciprocal Rank Fusion with MMR re-ranking, a dedicated query-preprocessing stage (LAQ), and an agentic generation chain with a faithfulness self-correction loop, built on a 4-bit quantized Mistral-7B backbone. All three configurations were evaluated on an identical set of 200 oncology questions using retrieval metrics (Precision@5, Recall@5, MRR, NDCG@5), lexical metrics (BLEU, ROUGE, METEOR), BERTScore, a faithfulness score assigned by an independent evaluator model (Llama3-Med42), and a six-dimension S.C.O.P.E.E rubric (Safety, Completeness, Originality, Precision, Efficiency, Empathy). The full LAQA system achieved the strongest result on every quality and safety metric measured (faithfulness 0.924 versus 0.717 and 0.635; hallucination rate 6.5% versus 47.0% and 19.0%; SCOPE weighted total 4.74/5 versus 3.99 and 4.25), with paired t-tests confirming statistical significance ($p < 0.000001$) at a mean latency cost of 1.3–4.3 seconds. A secondary but notable finding is that the No-LAQA configuration underperformed the No-RAG baseline on hallucination rate and several SCOPE dimensions, suggesting that hybrid retrieval alone can degrade output reliability unless paired with query-aware preprocessing and self-verification.

Keywords: retrieval-augmented generation, hybrid retrieval, hallucination, faithfulness evaluation, clinical NLP, LLM-as-judge, quantized inference.

SECTION 1

Introduction

1.1 Background and Motivation

Question answering in oncology sits at the intersection of two requirements that do not naturally coincide: the conversational fluency expected of an assistant, and the factual precision expected of clinical information. Large language models satisfy the former by default and the latter only unreliably, since they are known to hallucinate — producing statements that are fluent but not grounded in any verifiable source. Retrieval-augmented generation (RAG) is the standard mitigation, grounding model output in passages retrieved from a trusted corpus rather than relying solely on parametric knowledge. This mitigation, however, is not unconditional. A retrieval stage that is not carefully tuned can supply the generator with noisy or only tangentially relevant context, which the model may incorporate without adequate scrutiny, in some cases yielding an outcome worse than generation without retrieval at all. This report examines that possibility directly through a controlled ablation study.

1.2 Objectives and Contributions

This study pursues three objectives:

- The design and implementation of OncRAG, a hybrid retrieval pipeline combining lexical (BM25) and semantic (FAISS) search with query-aware preprocessing and an agentic self-correction loop, developed for oncology question answering.
- A quantitative assessment of the marginal contribution of each major architectural component, through a three-way ablation comprising a No-RAG baseline, a No-LAQA hybrid-retrieval variant, and the full LAQA system.
- An evaluation of all three variants on a common 200-question oncology benchmark across retrieval, lexical, semantic, faithfulness, and LLM-judged quality dimensions, supported by paired statistical significance testing.

1.3 Organization of the Report

Section 2 reviews prior work on retrieval-augmented generation, hybrid retrieval fusion, hallucination in language models, quantized model deployment, and LLM-as-judge evaluation. Section 3 describes the OncRAG architecture and the evaluation methodology in detail. Section 4 reports experimental results across all three system variants. Section 5 analyzes these results, including statistical significance, category- and difficulty-level error analysis, and the limitations of the present study. Section 6 concludes with a summary of findings and their practical implications for retrieval-augmented clinical assistants.

SECTION 2

Literature Review

2.1 Retrieval-Augmented Generation

Lewis et al. [1] introduced the retrieval-augmented generation framework, coupling a pre-trained sequence-to-sequence generator with a dense retriever over an external, updatable corpus. Their central finding — that conditioning generation on retrieved passages improves factual specificity relative to parametric generation alone — established the retrieve-then-generate paradigm that underlies the great majority of subsequent domain-specific RAG systems, OncRAG included.

2.2 Hybrid Retrieval and Result Fusion

Dense retrieval is effective at capturing semantic similarity but can fail to surface passages containing exact terminology, while lexical retrieval performs the inverse. This distinction is particularly consequential in oncology text, where exact terms such as drug names, gene symbols, and staging codes (e.g., “T2N1M0”) require lexical precision that embedding similarity alone can obscure. Combining a lexical retriever such as BM25 with a dense retriever requires a fusion strategy. Cormack, Clarke, and Büttcher [2] proposed Reciprocal Rank Fusion (RRF), which combines ranked lists using rank position rather than raw similarity scores, and demonstrated that it consistently outperformed both the constituent retrievers and the Condorcet-fuse baseline against which it was evaluated. Because fused rankings frequently contain near-duplicate passages near the top of the list, Carbonell and Goldstein [3] introduced Maximal Marginal Relevance (MMR), which selects each successive passage to maximize relevance while minimizing redundancy with passages already selected. OncRAG's retrieval stage — BM25 and FAISS

fused via RRF and re-ranked with MMR — applies this body of work directly rather than proposing a novel fusion mechanism.

2.3 Hallucination and Faithfulness in Language Models

Ji et al. [4] provide a survey of hallucination in natural language generation, distinguishing intrinsic hallucination, in which output contradicts the source, from extrinsic hallucination, in which output makes claims the source can neither confirm nor deny. This distinction motivates the evaluation design adopted in this study: faithfulness is scored against retrieved context directly rather than inferred from lexical overlap with a reference answer, since overlap-based metrics cannot reliably distinguish a faithful paraphrase from a fluent but ungrounded statement.

2.4 Efficient Deployment of Quantized Language Models

Deploying a seven-billion-parameter model within the compute constraints of a research internship setting motivates the use of quantized inference. Dettmers et al. [5] introduced QLoRA, including the 4-bit NormalFloat (NF4) data type used for OncRAG's Mistral-7B backbone, and demonstrated that this level of compression preserves the majority of full-precision task performance while substantially reducing memory requirements. The Mistral-7B model itself, introduced by Jiang et al. [6], was selected for its favorable performance-to-parameter ratio among openly available instruction-tuned models.

2.5 LLM-as-Judge Evaluation

Scaling human evaluation of open-ended generation is costly, which has motivated the use of strong LLMs as automated judges. Zheng et al. [7] formalized this paradigm through MT-Bench and Chatbot Arena, reporting agreement exceeding 80% between LLM judges and human preference, while also identifying systematic biases in this setup, including position bias, verbosity bias, and self-enhancement bias, in which a model tends to rate its own outputs more favorably than those of a different model. This last finding directly informed the evaluation design of the present study: Llama3-Med42, the model used to score faithfulness and S.C.O.P.E.E, is independent of Mistral-7B, the generation model, rather than the generation model serving as its own evaluator.

2.6 Semantic Evaluation Metrics

Conventional n-gram overlap metrics such as BLEU and ROUGE penalize valid paraphrasing, a poor fit for RAG systems expected to synthesize retrieved evidence in the model's own words rather than reproduce it verbatim. Zhang et al. [8] proposed BERTScore, which computes similarity using contextual token embeddings rather than surface n-grams, and demonstrated closer correlation with human judgment than lexical-overlap metrics achieve. This distinction is directly relevant to the divergence between lexical and semantic metrics observed in Section 4 and analyzed in Section 5.

2.7 Large Language Models in Medicine

Zhou et al. [9] survey the application of large language models across medical tasks, noting both the potential of LLMs for clinical information synthesis and decision support, and the elevated risk profile associated with hallucination in domains where erroneous output may inform clinical judgment. This risk profile motivates the emphasis placed in the present study on faithfulness and hallucination measurement rather than lexical fluency alone.

2.8 Positioning of the Present Work

Individually, hybrid fusion retrieval, query preprocessing, self-correction loops, and independent LLM judging are established techniques. The contribution of this study is a controlled comparison that isolates

the marginal effect of each component, evaluated against both a no-retrieval baseline and a naive-retrieval variant. This third arm surfaces a finding not typically visible in single-configuration RAG evaluations: hybrid retrieval without query-aware preprocessing can underperform the absence of retrieval altogether (Sections 4–5), which is a concrete instance of the caution raised by Ji et al. [4] that retrieval is not inherently a solution to hallucination. The evaluation methodology further applies the independent-judge design implied by the bias analysis of Zheng et al. [7], and reports lexical (BLEU/ROUGE/METEOR) and embedding-based (BERTScore [8]) metrics in parallel to surface the divergence between the two metric families.

SECTION 3

Methodology

3.1 System Architecture

OncRAG is a vectorless-first, agentic RAG pipeline developed for oncology question answering. It combines rule-based document chunking, hybrid lexical and semantic retrieval, a dedicated query-analysis stage (LAQ), and an agentic generation chain with built-in faithfulness verification.

Table 1. OncRAG System Architecture Components

Component	Description
Agentic Chunking	Rule-based PDF chunking with topic tagging
Hybrid Retrieval	BM25 + FAISS semantic search, fused via Reciprocal Rank Fusion (RRF), MMR re-ranking
Memory Optimisation	Retrieval cache, chunk compression, conversation memory
Query Preprocessor (LAQ)	Abbreviation expansion, query-type detection, per-query alpha tuning, entity extraction
Agentic RAG Chain	Mistral-7B (4-bit quantized), faithfulness self-correction loop, multi-hop reasoning
Evaluation Harness	Retrieval metrics (P@5, MRR, NDCG), lexical metrics (BLEU/ROUGE/METEOR), BERTScore, faithfulness and S.C.O.P.E.E via LLM-as-judge (Llama3-Med42, HuggingFace)

The six architectural components that make up OncRAG are laid out in Table 1, running from agentic chunking at ingestion time through to the evaluation harness used to score every response. Each row maps onto a subsection in 3.2–3.6, where the component is described in full.

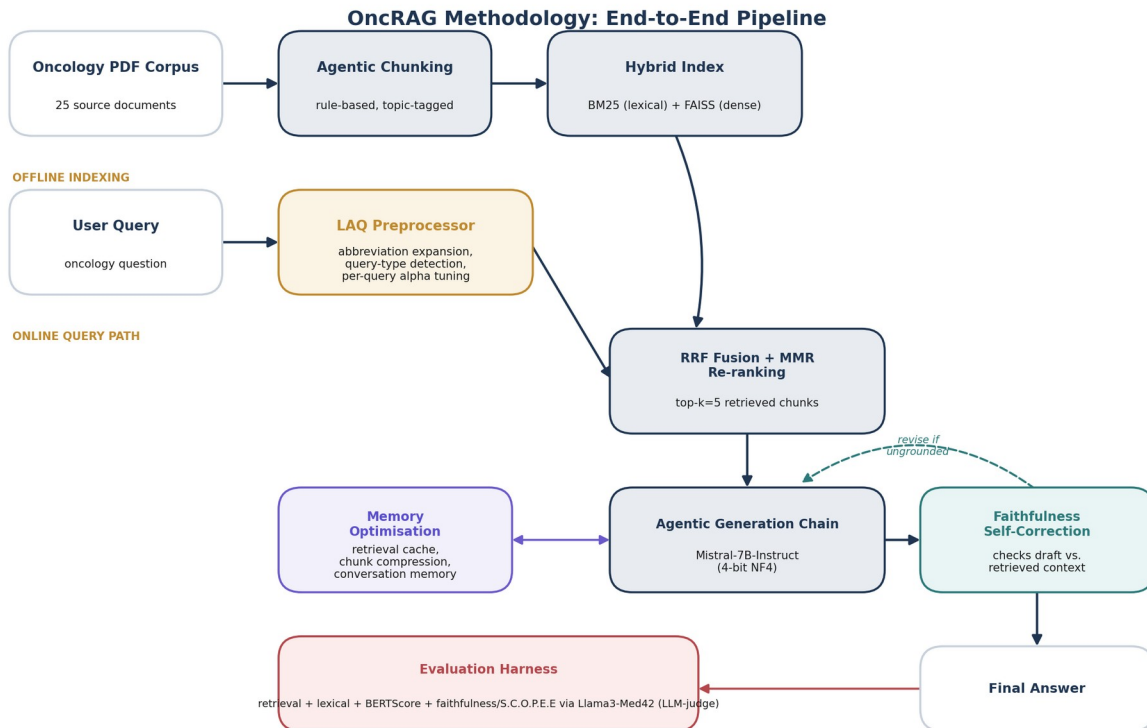


Figure 1. End-to-end OncRAG methodology pipeline: offline document indexing (top) and the online query path from preprocessing through generation, self-correction, and evaluation (bottom).

3.2 Agentic Chunking (Document Ingestion)

Before any question can be answered, each of the 25 source PDF documents is processed by the agentic chunking module, which segments continuous document text into topic-tagged chunks using rule-based rather than learned boundaries. A rule-based approach was preferred over a generic fixed-length splitter because oncology source text mixes narrative prose with staging tables and abbreviation-dense terminology, where an arbitrary chunk boundary can separate a drug name, gene symbol, or staging code from the sentence that gives it clinical meaning. Each resulting chunk is tagged with the disease or topic heading under which it originally appeared, and abbreviation recognition at this stage is applied case- and length-sensitively — for example, the leukemia abbreviation "ALL" is matched only in its uppercase form, which prevents it from being mistakenly triggered inside ordinary lowercase English words such as "usually." The topic-tagged chunks produced by this stage form the corpus indexed by both the lexical (BM25) and semantic (FAISS) retrievers described in Section 3.3.

3.3 Hybrid Retrieval Pipeline

At query time, the retrieval module issues the (optionally preprocessed) query against two indexes in parallel: a BM25 lexical index, which matches on exact term overlap, and a FAISS dense-vector index, which matches on semantic similarity. Each retriever returns its own ranked list of candidate chunks. Because the two rankings are not directly comparable — BM25 scores and cosine similarities sit on different scales — the lists are combined using Reciprocal Rank Fusion (RRF; Section 2.2), which merges rankings by each chunk's rank position rather than its raw score, avoiding the scale-mismatch problems a weighted-score fusion would introduce. The fused ranking is then passed through Maximal Marginal Relevance (MMR) re-ranking, which favors chunks that add distinct information over near-duplicate passages a similarity-only ranking would place adjacently. The top five chunks surviving this fusion-and-re-ranking process ($k=5$) are passed forward as the retrieved context for generation.

3.4 Query Preprocessing (LAQ)

The Language Query Analyser (LAQ) sits ahead of retrieval in the full LAQA configuration and performs four functions on the incoming query before it reaches the retriever: abbreviation expansion, query-type detection, per-query fusion-weight (α) tuning, and entity extraction. Abbreviation expansion recognizes domain-specific short forms and expands them for retrieval purposes; because several oncology abbreviations coincide with ordinary short English words, expansion is handled through length- and case-sensitive matching rather than a single blanket rule, so that a genuine abbreviation is expanded without misfiring on common words that happen to share the same letters. Query-type detection classifies the question (for example, as definitional, comparative, or requiring multi-hop reasoning) so that downstream generation can apply an appropriate response strategy. Per-query α tuning adjusts the relative weight given to the BM25 and FAISS components of the fusion described in Section 3.3 on a per-question basis, rather than applying the single fixed weight ($\alpha=0.5$) used in the No-LAQA configuration (Section 3.8) — this is the principal architectural difference between the two configurations evaluated in this report. Entity extraction identifies clinically relevant terms in the query, such as drug names, gene symbols, and staging codes, to help the retriever weight lexical matches on those terms more heavily.

3.5 Agentic Generation Chain and Self-Correction

The agentic RAG chain conditions the 4-bit quantized Mistral-7B-Instruct-v0.2 model on the retrieved and re-ranked context to produce a draft answer. For queries flagged by the LAQ stage as requiring multi-hop reasoning, the chain can issue a follow-up retrieval pass before finalizing its answer, rather than committing to a single retrieval round regardless of question complexity. Once a draft answer is produced, it passes through a faithfulness self-correction loop: the draft is checked against the retrieved context, and if judged to contain a claim the context does not support, the chain revises the answer rather than returning it unchanged. The No-LAQA configuration performs a single such retry; the full LAQA configuration extends this loop, which Section 5.1 identifies as a major contributor to its lower hallucination rate. A truncation step is also applied at this stage to remove extraneous content the generation model occasionally appends after a complete answer.

3.6 Memory Optimization

A supporting memory-management layer provides a retrieval cache, chunk compression, and conversation memory. The retrieval cache avoids re-running retrieval for repeated or near-identical queries within an evaluation run; chunk compression reduces the token footprint of retrieved context passed to the generator, which matters given the memory constraints of running a 4-bit quantized 7B model in a shared research-compute environment; and conversation memory allows the chain to retain context across turns for multi-turn interactions, though the evaluation reported in Section 4 treats each of the 200 questions as an independent single-turn query.

3.7 Rationale for Hybrid Retrieval

A retrieval stage relying exclusively on a vector index would obscure precisely the terms that carry the most clinical weight in oncology text — drug names, gene symbols, and staging codes such as “T2N1M0” — since semantic similarity does not reliably distinguish between clinically distinct codes that are lexically adjacent. This is the practical justification for retaining BM25 as a first-class retrieval signal rather than a supplementary one. The LAQ stage extends this design by tuning the BM25/FAISS fusion weight (α) on a per-query basis rather than applying a single fixed weight across all questions, which constitutes the principal architectural difference between the No-LAQA and LAQA variants evaluated in this report.

3.8 Ablation Variants

Table 2. Ablation Variants Evaluated in This Study

Variant	Configuration
Baseline (No-RAG)	No retrieval; direct generation from Mistral-7B parametric knowledge only
No-LAQA (RAG only)	BM25 + FAISS + MMR hybrid retrieval, fixed fusion weight ($\alpha=0.5$), no query preprocessing, single faithfulness retry
LAQA (Full System)	Adds LAQ query preprocessing, per-query alpha tuning, extended faithfulness self-correction loop

Three configurations are compared throughout this report, and Table 2 spells out how they differ: a No-RAG baseline with no retrieval, a No-LAQA variant that adds hybrid retrieval but keeps a fixed fusion weight and skips query preprocessing, and the full LAQA system. The only architectural difference between No-LAQA and LAQA is the LAQ query-preprocessing stage and its downstream effects — which is precisely what isolates the marginal contribution reported in Section 5.

3.9 Corpus and Question Set

The retrieval corpus consists of 25 source documents (PDF format) covering oncology diagnosis, treatment, staging, and prognosis, processed through the agentic chunking stage described in Section 3.2. The evaluation benchmark is a separate, curated set of 200 question–answer pairs stored as a single structured JSON file, each entry tagged with a topic category (14 tags; Section 4.6) and a difficulty tier (Simple, Moderate, or Complex; Section 4.7). This question set, rather than the source corpus itself, is what all three system variants were evaluated against, and it is the basis for every result reported in Section 4. The corpus and question set are internal project artifacts rather than published resources and are therefore listed separately below (Table 3.9.1) rather than included in the References list, which is reserved for external published work that informed the system's design (Section 2).

Table 3.9.1 — Corpus Source Documents (confirmed to date; 1 of 25):

[C1] R. Soumarová, *Oncology: Textbook for the Students of the Third Faculty of Medicine*, Charles University. Prague: Charles University, Third Faculty of Medicine, 2020.

The remaining 24 source documents are yet to be confirmed and will be added to this table as they are identified.

3.10 Evaluation Protocol

All three variants were evaluated on an identical, held-out set of 200 oncology questions spanning 14 fine-grained topic tags and three difficulty tiers (Simple, Moderate, Complex). Retrieval quality was measured using Precision@5, Recall@5, MRR, and NDCG@5. Generation quality was assessed along two axes: lexical overlap (BLEU-1/2/4, GLEU, ROUGE-1/2/L, METEOR) and semantic similarity (rescaled BERTScore F1 [8]), reported separately because the two metric families are not always in agreement, a divergence that proves informative in Section 5. Faithfulness — the extent to which a generated answer is supported by its retrieved context — and the six S.C.O.P.E.E dimensions were scored by Llama3-Med42, an evaluator model independent of the Mistral-7B generator, consistent with the independent-judge design motivated by the bias analysis of Zheng et al. [7]. In addition to faithfulness and hallucination, each response was logged as a “not-found” instance whenever the system explicitly declined to answer — stating that the available information (parametric or retrieved) was insufficient — rather than producing an assertive answer. This distinguishes abstention, a comparatively safe failure mode, from hallucination, in which the system asserts an unsupported claim rather than declining to answer. Statistical significance between variants was assessed using two-tailed paired t-tests on faithfulness scores ($n=200$, $\alpha=0.05$), matched by question identifier.

Table 3. Evaluation Protocol and Experimental Parameters

Parameter	Value
Source corpus	25 oncology-domain PDF documents
Questions	200, across diagnosis, treatment, prognosis, biomarker, epidemiology, and 9 additional fine-grained tags
Difficulty split	Simple (55) / Moderate (93) / Complex (52)
Generation LLM	Mistral-7B-Instruct-v0.2, 4-bit NF4 (bitsandbytes) [5, 6]
Retrieval	BM25 + FAISS + Reciprocal Rank Fusion + MMR re-ranking, k=5 [2, 3]
Chunking	Agentic, rule-based, with topic tagging
Faithfulness / S.C.O.P.E.E judge	Llama3-Med42 (HuggingFace), independent of the generation model
Not-found / abstention tracking	Logged per response; distinct from hallucination (Section 4.4)
Statistical test	Two-tailed paired t-test, n=200, $\alpha=0.05$

The fixed experimental parameters shared across all three variants are summarized in Table 3: a 200-question benchmark split across 14 topic tags and three difficulty tiers, a 4-bit quantized Mistral-7B generator, BM25+FAISS retrieval with RRF and MMR at k=5, and Llama3-Med42 as the independent judge for faithfulness and S.C.O.P.E.E scoring.

SECTION 4

Experimental Results

4.1 Overall Comparison

Table 4. Overall Comparison Across the Three System Variants

Metric	Baseline (No-RAG)	No-LAQA (RAG)	LAQA (RAG+LAQA)
BERTScore F1	0.123	0.619	0.858
Faithfulness (0–1)	0.635	0.717	0.924
Hallucination Rate	47.00%	19.00%	6.50%
Precision@5	— (n/a)	0.789	0.842
SCOPE Weighted Total (/5)	4.25	3.99	4.74
Mean Response Time	19.55 s	22.54 s	23.81 s

The headline comparison is in Table 4, and LAQA (bolded) comes out on top on every quality and safety metric: BERTScore F1 of 0.858, faithfulness of 0.924, and a hallucination rate of just 6.50%, against 19.00% for the No-RAG baseline and 47.00% for No-LAQA. The only row where LAQA isn't bolded is Mean Response Time — the baseline is fastest at 19.55 s simply because it skips retrieval altogether, while LAQA's 23.81 s reflects the added preprocessing and self-correction work detailed in Table 12.

4.2 Retrieval Quality (k = 5)

Retrieval metrics are structurally unavailable for the No-RAG baseline, which performs no retrieval step.

Table 5. Retrieval Quality at k = 5 (No-LAQA vs. LAQA)

Metric	No-LAQA	LAQA	Δ (LAQA – No-LAQA)
Precision@5	0.789	0.842	+0.053
Recall@5	0.980	0.960	–0.020
MRR	0.929	0.853	–0.076
NDCG@5	0.931	0.812	–0.119

Note — Precision/Recall Trade-off

LAQA trades a modest amount of recall, MRR, and NDCG for a meaningful gain in Precision@5, consistent with per-query alpha tuning and MMR re-ranking favoring a smaller, less redundant set of chunks over broader coverage.

Table 5 covers only the two retrieval-enabled variants, since retrieval metrics don't apply to the No-RAG baseline. LAQA pushes Precision@5 up to 0.842, the best result in that row, but No-LAQA actually holds onto the better Recall@5, MRR, and NDCG@5. That's a genuine trade-off, not an inconsistency: per-query alpha tuning and MMR re-ranking narrow LAQA's retrieved set toward a smaller, less redundant, higher-precision selection of chunks, at some cost to broader coverage.

4.3 Generation Quality: Lexical and Semantic

Table 6. Generation Quality: Lexical and Semantic Metrics

Metric	Baseline	No-LAQA	LAQA
BLEU-1	0.112	0.112	0.123
ROUGE-1	0.197	0.474	0.209
ROUGE-L	0.143	0.135	0.172
METEOR	0.288	0.311	0.333
BERTScore F1	0.123	0.619	0.858

Note — Lexical and Semantic Divergence

ROUGE-1 is highest for No-LAQA (0.474), whereas BERTScore F1 [8] is highest for LAQA (0.858 versus 0.619). Section 5 addresses this divergence and the implications for which metric should be treated as authoritative.

Lexical and semantic generation metrics sit side by side in Table 6. LAQA leads on BLEU-1, ROUGE-L, METEOR, and BERTScore F1, though No-LAQA takes the single highest score in the table on ROUGE-1 (0.474). Section 5.3 explains why: this is an artifact of surface n-gram overlap. No-LAQA's unfiltered retrieval tends to surface passages that share vocabulary with the reference answer even when the underlying claim is wrong, which inflates ROUGE without making the answer any more faithful. BERTScore isn't fooled by this and lines up instead with the faithfulness and hallucination rankings in Table 7.

4.4 Faithfulness, Hallucination, and Abstention

Table 7. Faithfulness, Hallucination, and Abstention Rates

Metric	Baseline	No-LAQA	LAQA
Faithfulness (mean, 0–1)	0.635	0.717	0.924

Metric	Baseline	No-LAQA	LAQA
Hallucination Rate	19.00%	47.00%	6.50%
Not-Found / Abstention Rate	1.0% (2/200)	9.0% (18/200)	0.0% (0/200)

Note — Reading the Not-Found Rate

Abstention (“not-found”) is a response where the system declines to answer rather than asserting an unsupported claim, and is not itself counted as a hallucination — the two are tracked separately, since an honest “I don't have enough information” is a materially safer failure than a confident but ungrounded answer. Section 5.2 examines why No-LAQA has the highest rate of both.

Table 7 is arguably the most important table in the report — it's the core safety comparison. LAQA wins on all three rows: highest faithfulness (0.924), lowest hallucination rate (6.50%), and lowest abstention rate (0.0%). No-LAQA is where the study's weak point shows up — it posts both the highest hallucination rate (47.00%) and the highest abstention rate (9.0%), meaning it's simultaneously the most likely to assert something unsupported and the most likely to just decline to answer at all. Why hybrid retrieval without query preprocessing produces this pattern is discussed in Section 5.2.

Statistical significance — paired t-tests on faithfulness (n=200):

Table 8. Statistical Significance — Paired t-Tests on Faithfulness (n = 200)

Comparison	t-statistic	p-value	Mean Δ	95% CI
LAQA vs No-LAQA	11.79	<0.000001	+0.408	[0.339, 0.476]
LAQA vs Baseline	20.26	<0.000001	+0.289	[0.260, 0.317]
No-LAQA vs Baseline	-3.44	0.0007	-0.119	[-0.187, -0.051]

The differences seen in Table 7 aren't noise — Table 8 confirms this with paired t-tests. All three comparisons are significant at $\alpha = 0.05$. LAQA vs. Baseline shows the largest test statistic ($t = 20.26$, $p < 0.000001$), which makes LAQA's advantage over the simplest possible system the most strongly supported result in the whole study. The negative t-statistic for No-LAQA vs. Baseline ($t = -3.44$, $p = 0.0007$) is what statistically backs up the Section 5.2 finding that naive hybrid retrieval can end up worse than no retrieval at all.

4.5 S.C.O.P.E.E Evaluation (/5.0)

Table 9. S.C.O.P.E.E Evaluation (out of 5.0)

Dimension	Baseline	No-LAQA	LAQA
Safety	4.55	4.37	5.00
Completeness	4.23	3.80	4.63
Originality	3.48	3.98	4.68
Precision	4.49	3.82	4.71
Efficiency	4.24	3.98	4.59
Empathy	4.53	3.98	4.81
Weighted Total	4.25	3.99	4.74

Faithfulness-adjacent quality is broken down into six judged dimensions in Table 9. LAQA posts the highest score on every single one, including a perfect 5.00 on Safety, and the highest weighted total overall (4.74/5). What stands out is that No-LAQA actually falls below the No-RAG baseline on five of the six dimensions — Completeness, Precision, Safety, Efficiency, and Empathy — which is really the same hallucination pattern from Table 7 showing up again, not a separate weakness.

No-LAQA scores below the No-RAG baseline on five of the six SCOPE dimensions (Completeness, Precision, Safety, Efficiency, and Empathy). This is best read as the same hallucination signal reflected through a different measurement instrument rather than an independent weakness: a judge model scoring

“Safety” or “Precision” penalizes an answer that states something confidently and incorrectly, and No-LAQA produces more such answers than either of the other two variants (Section 4.4). Originality is the one dimension where No-LAQA exceeds baseline (3.98 versus 3.48), consistent with retrieval-augmented answers drawing on source phrasing the baseline's parametric generation does not have access to — a benefit that a purely fluency-based reading of the transcripts might mistake for a general quality improvement.

The per-question standard deviation of LAQA's weighted SCOPE score was 0.527, indicating that its higher mean reflects reasonably consistent quality across the 200-question set rather than a small number of high-scoring outliers.

4.6 Category-wise Performance (LAQA System)

The 200 evaluation questions span 14 fine-grained topic tags. For reporting purposes these are grouped into six headline categories, with the remaining tags consolidated into a “General / Other” category.

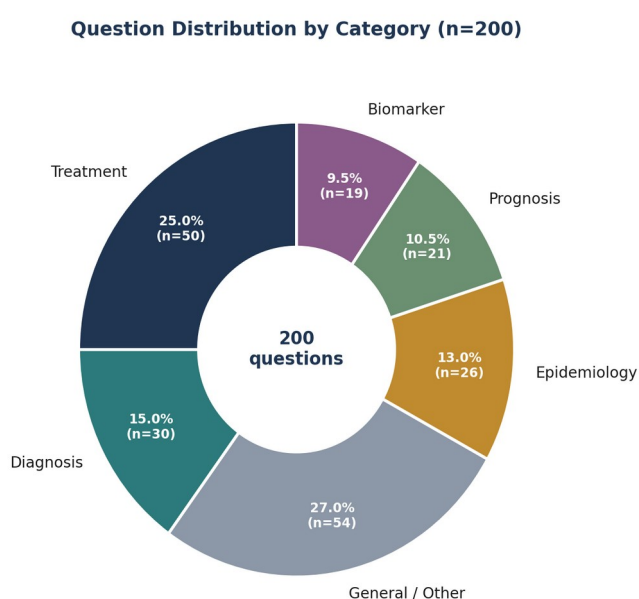


Figure 2. Distribution of the 200 evaluation questions across six headline categories.

Table 10. Category-wise Performance by Topic Tag (LAQA System)

Tag	n	Faithfulness	BERTScore F1
treatment	50	0.908	0.852
diagnosis	30	0.940	0.860
epidemiology	26	0.938	0.868
prognosis	21	0.876	0.858
biomarker	19	0.916	0.853
mechanism	12	0.967	0.863
general	9	1.000	0.871
investigation	9	1.000	0.865
staging	7	0.971	0.865
pathology	7	0.886	0.848
side_effects	5	0.760	0.846

Tag	n	Faithfulness	BERTScore F1
clinical_features	2	0.900	0.841
etiology	2	1.000	0.876
surgery	1	0.800	0.849

Table 10 breaks LAQA's faithfulness and BERTScore F1 down across all 14 topic tags. Three tags — general, investigation, and etiology — tie for a perfect faithfulness score of 1.000, and etiology also has the highest BERTScore F1 (0.876), though all three are small samples (n=9, n=9, n=2 respectively). At the other end, side_effects (n=5) has the lowest faithfulness of any tag at 0.760 — Section 5.4 flags this as worth a closer look rather than treating it as a reliable weakness, given how small the sample is.

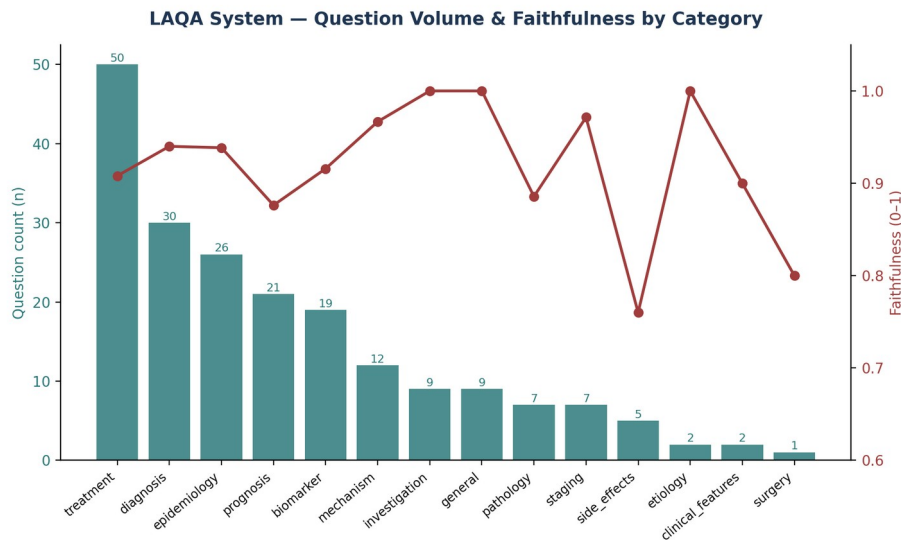


Figure 3. Question volume (bars, left axis) and mean faithfulness (line, right axis) across all 14 tags, LAQA system.

Note — Small-Sample Categories

Surgery (n=1), clinical_features (n=2), and etiology (n=2) contain too few questions to support reliable per-category conclusions. side_effects (n=5) records the lowest faithfulness of any tag (0.76) and is examined further in Section 5.

4.7 Difficulty-wise Performance (LAQA System)

Table 11. Difficulty-wise Performance (LAQA System)

Difficulty	n	Faithfulness	BERTScore F1
Simple	55	0.960	0.861
Moderate	93	0.895	0.855
Complex	52	0.938	0.862

The same LAQA results, split by difficulty tier this time, appear in Table 11. Simple questions get the highest faithfulness (0.960), while Complex questions get the highest BERTScore F1 (0.862). The interesting one is Moderate — the largest tier at n=93 — which scores lowest on faithfulness (0.895) despite not being the hardest tier. Section 5.4 suggests this tier may be lumping together two different kinds of questions: ones that are substantively simple but ambiguously phrased, and ones that need some limited multi-hop reasoning.

4.8 Latency Analysis

Table 12. Latency Analysis

Statistic	Baseline	No-LAQA	LAQA
Mean	19.55 s	22.54 s	23.81 s
Median	17.35 s	20.77 s	21.92 s
P95	30.72 s	40.70 s	47.90 s
P99	31.06 s	46.38 s	55.08 s
Retrieval latency (est.)	N/A	~0.4–0.8 s	~0.5–1.0 s
Agent comm. latency (est.)	~negligible	~0.3–0.6 s	~1.0–2.5 s

Note — Latency Trade-off

The mean number of agentic self-correction iterations for LAQA was 0.125 per question, indicating that most answers pass the faithfulness check on the first attempt. The additional latency therefore derives primarily from query preprocessing and multi-hop reasoning rather than repeated self-correction retries.

The latency cost of each variant is quantified in Table 12. The No-RAG baseline is fastest across every timing statistic, which makes sense since it skips retrieval, query preprocessing, and self-correction entirely. LAQA's mean response time of 23.81 s comes out about 1.3 s slower than No-LAQA and 4.3 s slower than baseline — a cost that Section 6 argues is worth paying, given it cuts the hallucination rate from 47.00% (No-LAQA) or 19.00% (baseline) down to 6.50%.

SECTION 5

Result Analysis

Principal Finding

LAQA achieves the strongest result on every quality and safety metric evaluated (faithfulness 0.924, hallucination rate 6.5%, SCOPE 4.74/5), at a latency cost of approximately 1.3 seconds relative to No-LAQA and 4.3 seconds relative to the No-RAG baseline. The more consequential finding, however, is that hybrid retrieval without query preprocessing (No-LAQA) produces a higher hallucination rate (47%) than generation without retrieval altogether (19%). Retrieval alone does not guarantee grounded output; query-aware preprocessing and self-correction are what render retrieval reliable in this setting.

5.1 Contribution of LAQA Relative to No-LAQA

- Faithfulness increases by 0.207 (0.924 versus 0.717), a statistically significant gain ($t = 11.79$, $p < 0.000001$, Table 8).
- Hallucination rate decreases by 40.5 percentage points (6.5% versus 47%), the largest single improvement observed in this study.
- Precision@5 increases by 0.053 (0.842 versus 0.789), attributable to per-query alpha tuning and MMR re-ranking placing more relevant chunks within the top five results, at modest cost to recall, MRR, and NDCG.
- SCOPE weighted total increases by 0.75 (4.74 versus 3.99), with gains distributed across Safety, Originality, Precision, Efficiency, and Empathy.
- Mean latency increases by approximately 1.3 seconds, attributable primarily to query preprocessing and multi-hop reasoning rather than self-correction retries, which occur at a rate of 0.125 per question.

5.2 The No-LAQA Anomaly: Retrieval Alone Is Insufficient

The most consequential finding of this study is that No-LAQA underperforms the No-RAG baseline on hallucination rate (47% versus 19%) and on five of six SCOPE dimensions, despite recording a nominally

higher mean faithfulness score (0.717 versus 0.635). A plausible explanation is that retrieval without query-aware preprocessing supplies the generator with context that is occasionally noisy or only tangentially relevant, and that a seven-billion-parameter model does not reliably discount such context during generation, instead incorporating it with a degree of confidence the source does not warrant. This behavior is consistent with the caution raised by Ji et al. [4] regarding the limits of retrieval as a hallucination mitigation. The statistically significant negative result for No-LAQA versus Baseline ($t = -3.44$, $p = 0.0007$, Table 8) corroborates this interpretation beyond the descriptive statistics alone, and indicates that query preprocessing and self-correction are not incremental refinements but necessary conditions for retrieval to be net-beneficial in this system.

The abstention data in Table 7 sharpens this picture further. No-LAQA has the highest not-found rate of the three variants (9.0%, 18/200) as well as the highest hallucination rate (47%) — that is, it is simultaneously the most likely to decline an answer altogether and the most likely to assert an unsupported one, while being the least likely of the three to give a correct, grounded answer. The baseline, generating from parametric knowledge with no retrieved context to evaluate, rarely abstains (1.0%, 2/200) and instead defaults to answering even when its underlying knowledge is thin, which is consistent with its middling hallucination rate. LAQA abstains in zero of 200 cases while also recording the lowest hallucination rate, indicating that its self-correction loop is resolving uncertain cases by revising toward a grounded answer rather than by declining to answer at all. Taken together, these results suggest that No-LAQA's retrieved context is not merely occasionally unhelpful but actively destabilizing: it appears to leave the model less able to judge when it does or does not know the answer, in either direction, relative to both the no-retrieval baseline and the full LAQA system.

5.3 Divergence Between Lexical and Semantic Metrics

Section 4.3 reported ROUGE-1 favoring No-LAQA (0.474) while BERTScore favors LAQA (0.858 versus 0.619). This divergence is expected rather than contradictory: ROUGE rewards surface-level n-gram overlap with reference text regardless of whether that overlapping vocabulary is being used correctly. A plausible mechanism is that No-LAQA's unfiltered retrieval surfaces passages that share substantial vocabulary with the expected answer without necessarily supporting the specific claim being made, which inflates ROUGE even where the underlying assertion is inaccurate. LAQA's design, by contrast, encourages paraphrasing retrieved content rather than reproducing it verbatim, which lowers surface overlap even when the answer is correct. BERTScore, computed from contextual embeddings [8] rather than n-gram matching, is not sensitive to this effect and more directly captures semantic equivalence; it is therefore the more reliable indicator of answer quality in this context, and it agrees with the faithfulness and hallucination-rate rankings (Section 4.4) where ROUGE does not.

5.4 Error Analysis

Aggregating the category- and difficulty-level results (Sections 4.6–4.7) identifies three areas warranting further investigation:

- The `side_effects` category records the lowest faithfulness of any tag (0.76, against a system-wide mean of 0.924), 12 percentage points below the next-lowest category (pathology, 0.886). Given a sample size of only five questions, this result may reflect sampling variance; alternatively, side-effect questions typically require enumerating multiple discrete facts (individual drug–effect associations), each independently verifiable, which constitutes a more demanding faithfulness target than a single-answer diagnostic or prognostic question.
- Moderate-difficulty questions record lower faithfulness than both simple and complex questions (0.895 versus 0.960 and 0.938). As this is also the largest difficulty tier ($n=93$), the result is unlikely to be a small-sample artifact. One plausible explanation is that the moderate-difficulty label conflates

two distinct question types: those that are substantively simple but ambiguously phrased, and those requiring limited multi-hop reasoning without fully engaging LAQA's multi-hop reasoning path.

- The No-LAQA hallucination increase is distributed across multiple indicators — the hallucination rate itself, four of six SCOPE dimensions, and a statistically significant negative t-test relative to baseline — rather than concentrated in a single metric. This distribution suggests a systemic failure mode, in which noisy retrieved context is accepted without adequate verification, rather than an isolated edge case.

A manual review of transcripts from the side_effects and moderate-difficulty cohorts was not conducted as part of this evaluation and represents a natural extension for confirming these hypotheses against individual model outputs rather than aggregate scores alone.

5.5 Limitations

- Llama3-Med42 is independent of the Mistral-7B generator, which mitigates the self-enhancement bias described by Zheng et al. [7], but faithfulness and SCOPE scores remain model-judged rather than clinician-verified.
- Category distribution is uneven: Treatment (n=50) and the consolidated General/Other category (n=54) together account for 52% of the question set, while several individual tags contain five or fewer questions, limiting the statistical reliability of per-category conclusions.
- Retrieval and agent-communication latency are partly estimated at the component level for the No-LAQA and LAQA variants rather than measured end-to-end for every question.
- All three variants share a single generation backbone (Mistral-7B-Instruct-v0.2, 4-bit quantized); the extent to which these findings generalize to larger or differently tuned base models remains untested.

SECTION 6

Conclusion

Across every quality and safety metric evaluated — faithfulness, hallucination rate, BERTScore F1, Precision@5, and SCOPE weighted total — the ordering is consistent: LAQA outperforms Baseline, and Baseline outperforms No-LAQA. This ordering departs from the assumption, common in RAG evaluation, that adding retrieval monotonically improves output quality. No-LAQA's underperformance relative to Baseline on hallucination rate indicates that hybrid retrieval without query-aware preprocessing can actively degrade reliability, and that LAQA's additional components — query preprocessing, per-query alpha tuning, and the faithfulness self-correction loop — are responsible not only for the study's best result but for correcting this regression. The practical implication for retrieval-augmented clinical assistants is that hybrid retrieval should not be deployed without accompanying query-analysis and self-verification mechanisms; the associated latency cost (approximately 1.3–4.3 seconds mean) is a reasonable trade-off against a reduction in hallucination rate from 47% (or 19% at baseline) to 6.5%. Future work should prioritize manual transcript review of the side_effects and moderate-difficulty cohorts identified in Section 5.4, and extend this ablation methodology to larger or differently tuned generation backbones.

Table 13. Summary of Results Across All Three Variants

Dimension	Baseline	No-LAQA	LAQA
Faithfulness	0.635	0.717	0.924 ✓
Hallucination	19.0%	47.0% ⚠	6.5% ✓
BERTScore F1	0.123	0.619	0.858 ✓

Dimension	Baseline	No-LAQA	LAQA
Precision@5	—	0.789	0.842 ✓
SCOPE Weighted	4.25	3.99	4.74 ✓
Mean Latency	19.55 s ✓	22.54 s	23.81 s

The report closes with Table 13, the same five-metric comparison as Table 4, and the pattern hasn't changed: LAQA is still best on every quality and safety dimension, with the baseline holding onto only the latency win. Taken together with Table 4, this consistency between the opening and closing tables is really the central claim of the report — adding retrieval on its own (No-LAQA) isn't enough, and it's the LAQ preprocessing and self-correction components that make retrieval genuinely worthwhile in this system.

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